

Visualizing the Spatio-Temporal Narratives of Hong Kong's Southbound Literati

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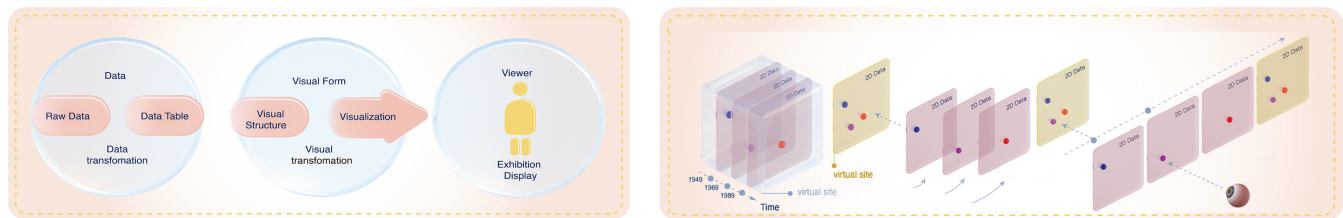


Figure 1: Process overview from corpus to floral visualization. Left: Card et al's three-stage visualization model [Card et al. 1999], illustrating the workflow from raw data and corpus preprocessing to visual encoding and exhibition display. **Right:** the translation of cleaned datasets into visual form using multiple encoding techniques, including time flattening, juxtaposition, and spatial-temporal mapping that culminate in the generation of data flower structures.

Abstract

Nanlai Zhi Hua visualizes the spatio-temporal narratives of Hong Kong's Southbound Literati through "data flowers" that merge biographical and fictional geographies. Integrating literary geography with information visualization, the project translates displacement

and nostalgia into visual form, revealing how spatial memory structures diasporic creativity in post-1949 Hong Kong.

CCS Concepts

• Human-centered computing → Visualization application domains; • Applied computing → Media arts; Literature.

Keywords

narrative visualization, literary geography, spatial humanities, diasporic literature, affective computing, digital humanities, exhibition display, cultural identity, geospatial narrative, media arts

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1 INTRODUCTION

The post-1949 migration of Chinese literati to Hong Kong formed the Southbound Literati, a diaspora whose works reflect “cultural bereavement” [Bhugra and Becker 2005]: disorientation, nostalgia, and loss of coherence. While their cultural displacement is well studied, its spatial dimension is not. This project applies literary geography and information visualization, extending Piatti [Piatti et al. 2013] and Card [Card et al. 1999], to reveal how real and imagined geographies intertwine in the works of Eileen Chang and Jin Yong, showing how spatial memory shapes literary expression in exile.

2 METHODOLOGY

Fictional spaces are fluid, shaped by narrative and reader interpretation. Building on this, we apply a three-stage transformation model—data transformation, visual encoding, and exhibition presentation—based on Card’s visualization framework [Card et al. 1999]. The visual encoding adapts spatiotemporal models by Andrienko et al. [Andrienko et al. 2003] and Bach et al. [Bach et al. 2017]. Our central metaphor, the flower, draws on Chinese aesthetics where floral imagery signifies growth and creativity. Each of the 25 “data flowers” encodes an author’s literary path, expressing both rootedness and displacement that mirror the Southbound Literati’s exile and creative renewal in post-1949 Hong Kong.

3 VISUALIZATION FRAMEWORK

3.1 Data Transformation

The initial stage centered on compiling and preprocessing biographical and narrative data from 25 representative Southbound authors. The corpus included fictional novels, autobiographical essays, and serialized texts emphasizing displacement and identity, selected according to literary prominence, geographic richness, and archival availability. For instance, Jin Yong’s revised works encompass over fifteen novels totaling more than ten million characters. To address disparities in text length, excerpts were normalized for thematic and geographic density. Lexical and spatial analyses, conducted using Voyant, WordTree, Excel, and MiniTagCloud, revealed recurring motifs such as cultural identity anxiety, displacement, and attachment to one’s hometown. Each textual dataset was parsed to extract two categories of spatial data: *lived locations*, referring to places where authors resided, worked, or traveled, and *literary locations*, representing sites embedded in their narratives. Manual validation excluded metaphorical or ambiguous terms (e.g., “mountain of sorrow”), retaining only spatially or narratively significant references. The cleaned datasets were integrated into Gephi through unified CSV files, where color distinguished lived from literary spaces, dot size indicated frequency of mention or visitation, and line thickness represented the strength of narrative or biographical connections. The ForceAtlas and YifanHu algorithms optimized spatial layout, revealing clusters around emotional landmarks and correlations between lived and imagined geographies. To preserve both legibility and interpretive detail, selective label disclosure was applied during visualization.

3.2 Visual Transformation

In the second stage, the cleaned datasets were transformed into visual form using color-time flattening, temporal juxtaposition, and generalized cube operations, following the methods of Andrienko et al. [Andrienko et al. 2003] and Bach et al. [Bach et al. 2017] (Figure 1). Color-time flattening compressed multiple temporal layers into a single two-dimensional map, displaying shifts in location, visit frequency, and key events over time. Temporal juxtaposition arranged lived and literary trajectories side by side, while node-link diagrams visualized spatial relationships and recurring motifs across works.

Informed by Bertin’s theory of visual variables [Bertin 1967], color, size, position, and line weight were used to encode spatial typology, frequency, and relational strength. The resulting “data flowers,” rendered with Houdini, Blender, and After Effects, visualized each author’s narrative evolution through layered petals and spatial motion. The compositions reveal striking convergence between lived and literary spaces; for instance, in Jin Yong’s visualization, recurrent imagery such as the Qiantang River tide and the misty rain over South Lake underscores how spatial nostalgia structurally shapes his creative imagination.

4 CONCLUSION

Nanlai Zhi Hua demonstrates the potential of information visualization to advance literary scholarship while preserving interpretive nuance. The data-flower metaphor effectively bridges computational analysis and humanistic inquiry, maintaining the openness of literary interpretation while revealing underlying spatial patterns within diasporic texts. The project establishes a methodological framework for analyzing the spatial manifestations of cultural displacement, providing transferable tools for the study of other migratory literary traditions, while future work employs interactive features such as gesture interaction, which will be presented through projection mapping and spatial design to a broader audience, enabling its accessibility as an immersive public experience.

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